Kamil Belán

Curriculum vitae

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kamilbelan

Education

2026–now **doctoral**, *Faculty of Mathematics and Physics, Charles University (CUNI FMP)*, Prague, Computational Mathematics
ongoing

2024–2026 **master**, *CUNI FMP*, Prague, Mathematical and Computational Modelling in Physics

2021–2024 **bachelor**, *CUNI FMP*, Prague, Physics

2017–2021 **grammar school**, Třebíč

Research interests

Numerical methods for partial differential equations of fluid dynamics; past work includes

- stratified compressible atmospheric flow
- Lagrangian (particle) methods, in particular Smoothed Particle Hydrodynamics (SPH)
- well-balanced numerical schemes

In the ongoing doctoral study the main research interests are higher order methods, in particular

- discontinuous Galerkin method (DGM)
- *hp*-adaptive discontinuous Galerkin method

Theses

2026 **Numerical methods for the simulation of air flow past a mountain**, *master's thesis*, supervisor doc. RNDr. Michal Pavelka, Ph.D.
Objective of the theses is to further develop and verify the SPH method for compressible atmospheric flow around orography. The work relies on a reformulation of the problem in natural variables, robust terrain-following orography description and improved boundary conditions treatment at the inflow and outflow. Hydrostatic (in)stability is addressed by the development of novel well-balanced schemes and adaptive techniques.

Codebase (both theses) `https://github.com/kamilbelan/sph-mountain-waves`

2024 **Simulating flows past a mountain range using smoothed particle hydrodynamics**, *bachelor's thesis*, supervisor doc. RNDr. Michal Pavelka, Ph.D.
The main objective of the thesis is to use the Lagrangian numerical method SPH to simulate compressible atmospheric flow over mountain orography, a problem typically solved by mesh methods (FVM, FD). It was shown by numerical evidence that the simulation of hydrostatic balance of a stratified atmosphere by the standard SPH method is so unstable, that SPH in its standard formulation is completely unable to capture phenomena like internal gravity waves.

Technical skills

Programming languages: Julia, Python, Mathematica, MATLAB

Tools: LaTeX, Git, SSH, bash, SLURM, Docker

Languages

Czech: native

English: **C1** – Cambridge certificate

German: ~**B1** – courses at Charles University